Bo Li

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Research Interests

AI-enabled chemical discovery; quantum chemistry, statistical mechanics, and molecular simulation; chemical knowledge and coordination equilibria for critical-mineral recovery; generative molecular design and reaction networks for reversible CO_2 capture; agentic screening and collective transport in disordered ionic materials; molecular mechanisms in catalytic conversion; interpretable machine learning and adaptive simulation.

Academic Appointments

Research Assistant Professor, Chemical and Biomolecular Engineering, Vanderbilt University — Jan. 2023 – present

Postdoctoral Research Associate, Theoretical Division, Los Alamos National Laboratory — Jan. 2020 – Jan. 2023

Postdoctoral Scholar, Department of Chemistry, University of California, Riverside — Nov. 2016 – Jan. 2020

Education

Ph.D., Theoretical Chemistry — 2015
Technical University of Munich, Germany. Advisor: Prof. Dr. Notker Rösch.
Computational Modeling of Chemical Systems in Solution: A Microscopic Approach for Efficient Density Functional Calculations with ParaGauss

M.Eng., Material Physics and Chemistry, Graduate University of Chinese Academy of Sciences, China — 2011

B.Sc., Geophysics, Peking University, China — 2008

Research Experience

Vanderbilt University — Jan. 2023 – present

- Lead research connecting electronic structure, intermolecular interactions, statistical mechanics, and transport in ionic liquids, molten salts, coordination compounds, and disordered materials.
- Develop generative molecular design for CO_2-reactive carbanions and interpretable, data-efficient ML models; integrate ab initio molecular dynamics, polarizable and machine-learned potentials, and enhanced sampling.
- Develop the REEL-QUEST chemical knowledge base and prototype agentic reaction workflows for chemically grounded discovery.
- Collaborate with experimental researchers to interpret spectroscopy, structural characterization, adsorption, and transport measurements.

Los Alamos National Laboratory — Jan. 2020 – Jan. 2023

- Developed density-functional tight-binding parameters for f-block chlorides and integrated DFTB with machine learning to enable larger molecular dynamics simulations.
- Modeled molten-salt and f-element systems for nuclear energy applications; investigated molten lanthanide chloride structures with ab initio molecular dynamics.

University of California, Riverside — Nov. 2016 – Jan. 2020

- Investigated uranyl binding with amidoxime ligands using molecular dynamics and enhanced free-energy sampling for seawater uranium extraction.
- Developed polarizable models of molten salts; applied DFT, MD, and QM/MM to CO_2 capture and electrocatalysis; contributed to DOE-NEUP proposals and technical reports.

Selected Publications

Across CO_2 capture, catalysis, coordination chemistry, and ionic transport. All 51 papers are on Google Scholar.

1. **Li, B.**; Jiang, D.-e. "Discovering CO_2-Reactive Carbanions via Property-Guided Generative AI." *J. Chem. Inf. Model.* **2026**, 66, 6983–6991.
2. **Li, B.**; Fu, Y.; Yang, Z.; Dai, S.; Jiang, D.-e. "Intermolecular Proton Transfer Enabled Reactive CO_2 Capture by the Malononitrile Anion." *J. Phys. Chem. B* **2024**, 128, 10207–10213.

3. Qiu, L.; **Li, B.**; Hu, J.; Ganesan, A.; Pramanik, S.; Damron, J. T.; Li, E.; Jiang, D.-e.; Mahurin, S. M.; Popovs, I.; Steren, C. A.; Fan, J.; Yang, Z.; Dai, S. "Cascade CO_2 Insertion in Carbanion Ionic Liquids Driven by Structure Rearrangement." *J. Am. Chem. Soc.* **2024**, 146, 29588–29598.
4. Zhu, C.; **Li, B.**; Li, C.; Lu, L.; Li, H.; Yuan, X.; Kang, X.; Jiang, D.-e.; Zhu, M. "Arylation of Gold Nanoclusters and Insights into Structure-related CO_2 Reduction Reaction Performances." *Chem. Sci.* **2025**, 16, 10273–10281.
5. Pramanik, S.; **Li, B.**; Driscoll, D. M.; Johnson, K. R.; Evans, B. R.; Damron, J. T.; Ivanov, A. S.; Jiang, D.-e.; Einkauf, J.; Popovs, I.; Jansone-Popova, S. "Tetradentate Ligand's Chameleon-Like Behavior Offers Recognition of Specific Lanthanides." *J. Am. Chem. Soc.* **2024**, 146, 25669–25679.
6. Lei, M.; **Li, B.**; Liu, H.; Jiang, D.-e. "Dynamic Monkey Bar Mechanism of Superionic Li-ion Transport in $LiTaCl_6$." *Angew. Chem. Int. Ed.* **2024**, 63, e202315628.

Teaching and Mentorship

- Developed and deployed computational infrastructure for a molecular simulation course for senior undergraduate and graduate students.
- Organized high-performance computing and molecular simulation workshops for chemistry students.
- Mentored graduate, undergraduate, and high-school researchers in computational chemistry, molecular simulation, and scientific programming.
- Teaching interests: chemical engineering thermodynamics, separation processes, and computational methods, as well as molecular thermodynamics, statistical mechanics, materials chemistry, and scientific AI.

Funding and Proposal Experience

Contributed computational strategy, technical sections, simulation plans, and HPC justification to multi-investigator proposals supported by DOE Basic Energy Sciences and DOE Nuclear Energy, including NEUP. Experience developing research concepts, revising proposals, and responding to reviewer feedback.

Presentations

Feb. 2024	Poster, Gordon Research Conference on Atomically Precise Nanochemistry, Galveston, TX.
Oct. 2023	Invited talk, Materials Science & Technology Conference, Columbus, OH.
Feb. 2023	Poster, Theory Frontiers in Actinide Science: Chemistry & Materials, Santa Fe, NM.
Jul. 2022	Poster, 19th Chinese-American Kavli Frontiers of Science Symposium, Irvine, CA.
Feb. 2020	Oral, International Workshop on Theory Frontiers in Actinide Sciences, Santa Fe, NM.
Aug. 2019	Oral, ACS Fall National Meeting, San Diego, CA.
Jun. 2019	Oral, ANS Annual Meeting, Minneapolis, MN.
May 2015	Poster, Bunsentagung, Bochum, Germany.
Mar. 2014	Oral, TUM-A*STAR Workshop on Computational Catalysis, Garching, Germany.

Professional Service

Journal reviewer: *Journal of Materials Chemistry C*, *The Journal of Physical Chemistry*, *Dalton Transactions*, *Nanoscale*, *Small Methods*, *ACS Applied Energy Materials*, *Applied Surface Science*, *Journal of Molecular Liquids*, *Advanced Science*, *Chemical Engineering Journal*, *Journal of Solution Chemistry*, and *Microporous and Mesoporous Materials*.

Reviewer of high-performance computing allocation proposals at Los Alamos National Laboratory. Member of the American Chemical Society, American Nuclear Society, and American Ceramic Society.

Honors and Awards

TUM International Graduate School of Science and Engineering Scholarship (2012); NIMTE Outstanding Student Scholarship (2010); CAS Graduate Student National Scholarship (2008).

Computational Expertise

Electronic structure and simulation: CP2K, VASP, Quantum ESPRESSO, DFTB+, ORCA, LAMMPS, GROMACS, AMBER.

Machine learning: PyTorch, TensorFlow, scikit-learn, Chemprop, Gaussian process regression, neural-network potentials, symbolic regression, diffusion models, and variational autoencoders.

Programming and HPC: Python (NumPy, SciPy, Pandas, ASE), Fortran, C/C++, Bash, MPI, CUDA, SLURM/PBS, Linux.